

Forest restoration effects on soil preferential flow in the paleo-periglacial eastern liaoning mountainous regions, China

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ARTICLE INFO

Handling Editor: Jian Zuo

Keywords:

Preferential flow development degree
Flow morphological characteristics
Solute migration
Dual-tracer technology
Forest restoration approaches
Paleo-periglacial landform

ABSTRACT

To provide a practical reference index for the protection of water conservation forests and the establishment of vegetation in different forest restoration areas, we seek to clarify the preferential flow development and its influencing factors under different rainfall events in the forest restoration areas of paleo-periglacial landform in eastern Liaoning Province of China. To quantify the preferential flow development (*PFI*) in soil profile-scale through soil profile indexes, we carried out dual-tracer experiments with dyeing in two types of forest restoration lands (plantation forest, PF, and natural secondary forest, NSF) in the paleo-periglacial landform forestland to obtain the soil water infiltration trajectories and water-solute transport characteristics. The indexes of preferential flow morphology and pathway diversity (solute migration) were integrated to construct a more complete evaluation system of the *PFI*. In addition, we explored factors affecting the *PFI* through soil physical properties and root characteristics. The results showed that: (1) There were obvious preferential flows in both PF and NSF. Compared with NSF, preferential flow in PF diverged earlier, had greater non-equilibrium degree, more paths and higher development degree, and preferential flow was the main infiltration form. (2) Changes in infiltration amount played a key role in altering the *PFI*, i.e., increasing infiltration amount reduced the *PFI*, especially in PF ($p < 0.05$); (3) After water infiltration, the solute Br^- diffusion range of the soil profile was wider than that of Brilliant Blue tracer, thus the development of preferential flow was better synthetically reflected by the indexes of water flow morphology and solute distribution characteristics; (4) Differences in the *PFI* can be explained by variations in clay content, total porosity (*TP*), and root volume density (*RVD*), with total path coefficients of 0.824, 0.462, and 0.624, respectively. In addition to establishing different forest restoration types in similar areas, it would be worthwhile to strengthen the prevention and control measures of soil and water loss in NSF, and properly manage PF in a near-natural model to enhance ecologically sustainable development of forest land. The results of this study contribute to the understanding of the hydrological processes of forest ecosystems in paleo-periglacial landforms, and provide theoretical support for the proposed forest management strategies.

1. Introduction

In the context of global warming, the frequency, intensity, and duration of extreme events have increased (Easterling et al., 2000; Allan and Soden, 2008), which has seriously affected the sustainability of forest ecosystem functions. Forest ecosystems provide important ecological services, including soil and water conservation, infiltration

promotion and flow reduction, and environmental improvement (Deng et al., 2023). It is well established that ecosystem function depends on the availability of soil moisture (Denissen et al., 2022), showing that soil water is of great significance in forest ecosystems. Concurrently, forest vegetation regulates rainfall redistribution and influences soil water content, which in turn affects soil hydrological processes (Demir et al., 2022). In addition, studies have shown that precipitation is spatially

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<https://doi.org/10.1016/j.jclepro.2024.142974>

Received 28 February 2024; Received in revised form 19 May 2024; Accepted 20 June 2024

Available online 22 June 2024

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more complex and temporally heterogeneous compared to other climate factors, such as temperature, at global and regional scales (Coumou and Rahmstorf, 2012; Xu et al., 2019), which undoubtedly complicates the prevention and treatment of forest hydrological impairment. This affects the soil infiltration that is an important part of the hydrological cycle and runoff formation process (Duan et al., 2021), as well as a key factor in constraining runoff generation and vegetation growth (Alaoui et al., 2011a).

Preferential flow is a common form of soil infiltration defined as inhomogeneous flow or non-equilibrium flow of fluids, gases, and solutes in preferential pathways through porous medium (Gerke, 2006; Lin, 2010). Soil with abundant preferential flow paths promotes rapid water flow to deeper soil, which means modified hydrological processes in the soil (De Boever et al., 2016; Dai et al., 2020a). Although spatially the preferential flow pathway represents only a portion of the total soil volume, preferential flows contribute 11%–94% of the overall infiltration amount (Alaoui, 2015). Considering the preferential flow characteristics, preferential flow potentially accelerates the flow of contaminants to groundwater, thereby threatening groundwater quality (Clothier et al., 2008; Gerke et al., 2010). Preferential flow is, however, positive for many ecosystem services, such as flood protection, erosion control, and aquifer recharge, despite their negative impacts on water quality, wastewater treatment, and food security (Clothier et al., 2008). Due to its importance, soil preferential flow has become a key issue in the evaluation of ecosystem services (Beven and Germann, 2013; Zhang et al., 2016). Thus, preferential flow parameters are often used as indicators for evaluating pathway morphology of water transport, including dye coverage, depth of diffusion area, percentage of preferential pathways, length index, stained pathways width, etc. (Van Schaik, 2009; Weiler and Flühler, 2004). In recent years, with the continuously improved understanding of preferential flow mechanisms under different land use, the preferential flow is often calculated by multiple indexes by using the above mentioned parameters (Zhang et al., 2017). There are also academics based on the hydrological connectivity of soil profiles to evaluate the developmental characteristics and the degree of the preferential flow (Dai et al., 2020a, 2020b). In addition, preferential flow pathways in vertical soil profiles can affect water redistribution and solute diffusion (Zhang et al., 2017). In addition to indicators of preferential flow morphology, solute distribution characteristics can also characterize preferential flow pathways distribution to a certain extent. However, the distributional characteristics of solute migration and the degree of preferential flow development in soil profiles based on the α -biodiversity theory have not been widely evaluated. Therefore, to evaluate the degree of preferential flow development using multi-indicator method that integrate the morphological and distribution diversity characteristics of preferential flow pathways can potentially improve the understanding of preferential flow development in different ecosystems.

In forest ecosystems, soil properties and root traits in the rhizosphere soil layer are key factors in inducing the generation of preferential flows. The soil macropore structure of soil properties has been widely recognized as an important system affecting hydrological processes. Soil macropores play a crucial role in the formation of the soil's loose and porous medium structure, also functioning as important conduits for soil water and gas transport and exchange (Beven and Germann, 1982). Soil macropores can reduce surface runoff by promoting soil water infiltration. The macroporous channels also facilitate the transport of solute-carrying water bypassing the soil matrix for rapid transport to deeper soil layers, and result in the formation of a common type of preferential flow known as macropore flow. In addition, other soil properties such as soil bulk density (BD), initial soil water content (ISWC), etc. Have been shown to have a significant impact on water infiltration (Molina et al., 2019; Schwen et al., 2011). Roots can change soil volumetric water content by absorbing water, increasing soil water storage capacity, and thus enhancing the ability of the forest land to conserve soil and water (Liu et al., 2015). Furthermore, the vegetation

root network can trigger preferential flow, which further affects the wetting front velocity and soil moisture response time to rainfall (Sidle et al., 2001). As for root system architecture, including root diameter, length, growth direction, topology structure, curvature etc., affect the formation of root channels, which in turn changes the preferential flow movement. Particularly if the root diameter is smaller than the root pore diameter, it will provide a potential cylindrical space for water transport (Ghestem et al., 2011). Therefore, it is crucial to delve into the role of soil properties and root traits in the rhizosphere soil layer of forest ecosystems that influences the development of preferential flow, to comprehend hydrological processes.

The periglacial landform is a type of landform developed by freeze-thaw action in non-glaciated cold regions encompassing approximately 20% of the global continental area (Gutiérrez and Gutiérrez, 2016). This landform type is subject to the Quaternary glacial periglacial force to form a variety of gravel widely distributed periglacial landforms, which is prone to major geological disasters such as mudslides. The Laotudingzi National Nature Reserve in eastern Liaoning Province of China is a typical paleo-periglacial landform type. Moreover, as an important water source in Liaoning Province, the region features numerous natural secondary forests (NSF) as well as plantation forests (PF), aiming to fulfill the functions of water conservation, provision of high-quality water for downstream cities, and soil erosion prevention. Our previous studies have shown that the extensive distribution of gravel and vegetation roots in the rhizosphere layer of the study area affects the water infiltration process, and water infiltration is dominated by preferential flow (Wang et al., 2024). Currently, with the implementation of the United Nations Decade on Ecosystem Restoration by the United Nations Environment Programme (2019), forest restoration is being widely implemented globally and has far-reaching multiple environmental and social impacts (Chazdon and Brancalion, 2019; Edwards et al., 2021). PF and NSF are the major means of forest restoration in ecosystems at present. Nevertheless, the characterization and evaluation of the development and degree of preferential flow in different forest restoration approaches of typical paleo-periglacial landform have not been widely investigated. Especially, the impact of the current large-scale establishment of PF on forest hydrological processes, meanwhile on preferential flow transport processes that include soil water movement and solute migration, is not clear.

Considering the importance of preferential flow in forest hydrological processes, it may be stated that there is a general lack of research on preferential flow mechanisms in different forest restoration approaches especially for of paleo-periglacial landforms. Accordingly, the purpose of this study was to: (1) Evaluate the characteristics of preferential flow under different rainfall conditions in different forest restoration approaches for the paleo-periglacial landform in eastern Liaoning Province of China; (2) Based on the characteristics of water and solute transport in the infiltration process, a multi-indicator method that integrates the morphology and pathways diversity indicators of preferential flow was investigated for the evaluation of preferential flow development; and (3) Improving the understanding of soil physical property and root traits mechanisms on the development of preferential flow. The findings can provide improved strategies for the establishment of local water conservation forests and the protection of regional water security.

2. Materials and methods

2.1. Site description

The study area is located in the Laotudingzi National Nature Reserve of Liaoning Province (41°11'11"–41°21'34"N, 124°41'13"–125°05'15"E), China (Fig. 1). It is a typical forestland of paleo-periglacial geomorphic area belonging to the northern temperate continental monsoon climate, with annual average temperature of 6.0 °C and absolute minimum temperature of −33.2 °C. It is the wettest area in Liaoning Province with an average annual precipitation of 827.8 mm concentrated in

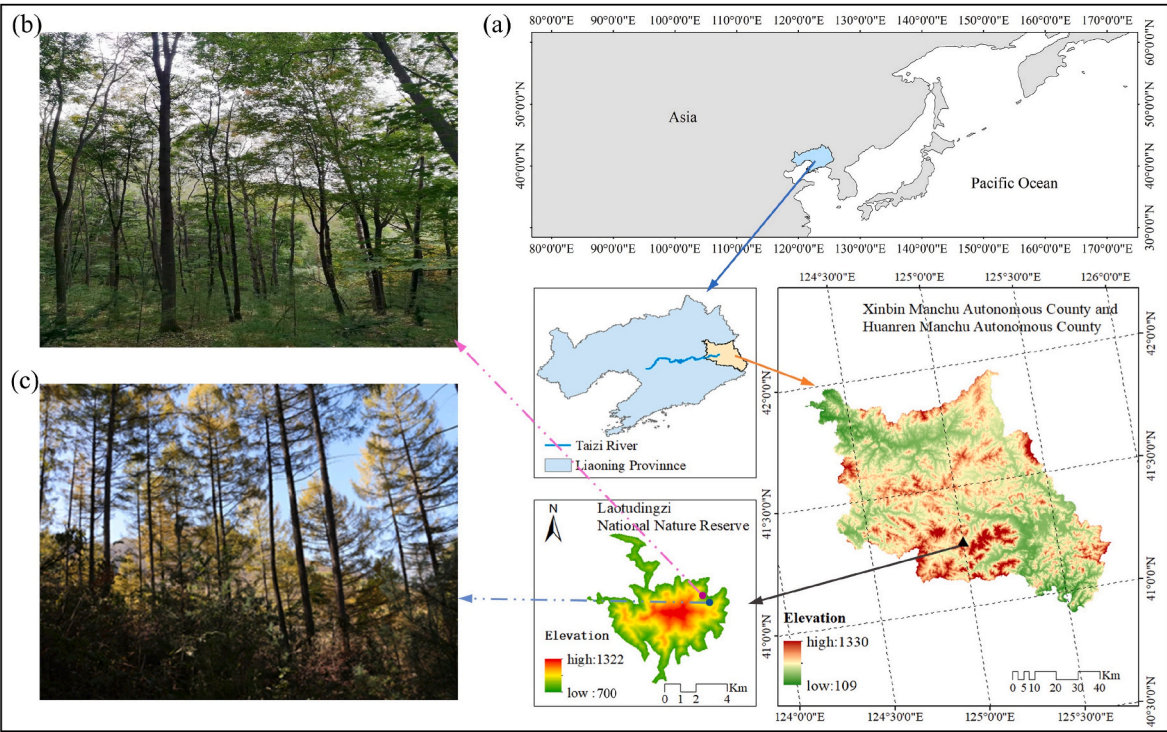


Fig. 1. Geographical location of the two investigated forest restoration sites. (a) Geographical location of the study area. The two approaches of forest restoration included (b) natural secondary forests (NSF) and (c) plantation forests (PF).

June–August. The main soil texture class is sandy loam (WRB, 2015). The forestland soil in the study area has high moisture content and adequate nutrients, which makes it easy for vegetation to grow, with many plantations and natural secondary forests established in this region. However, due to the special characteristics of the paleo-periglacial geomorphology, the gravel does not sequentially overlap the rhizosphere soil layer resulting in a barren and thin soil layer with poor stability. Thus, the vegetative roots are unstable and very prone to soil erosion.

The study area is the upper parts of the Taizi River (Fig. 1 (a)). As the industrial and agricultural production area of Liaoning Province, the Taizi River Basin accounts for about 70% of the province’s water consumption. However, due to the imbalance between supply and demand, the province is experiencing a serious water shortage. Therefore, improving the water security of the region is an urgent problem to be solved. A way in this direction is the establishing of water conservation forests, which are meant to improve ecosystem services and prevent soil erosion.

We selected two typical forest restoration areas (including plantation forests, PF, and natural secondary forests, NSF), which are widely distributed in the study area. To decrease errors and uncertainties, we selected the sample plots based on representativity in terms of elevation, slope, slope direction, and forest age as shown in Table 1.

2.2. Dual-tracer experiment

Dual tracers were selected for the infiltration experiments, including Brilliant Blue and NaBr. Brilliant blue is a widely used dye tracer due to

its color that can be diverged from the soil color and efficiently reflect water transport pathways. However, due to adsorption, it can attach to the soil particles, and its ability to migrate in the soil is affected to a certain extent. In contrast, Br⁻, although colorless and unable to visualize solute migration pathways, has the advantages of being soluble in water, having low soil background concentration, and low adsorption, which allows it to better characterize solute migration in soil. By combining NaBr and Brilliant Blue a truer visualization of solute transport can be achieved. The experimental rainfall for infiltration experiments was selected to be 35 mm (heavy rainfall) and 60 mm (extreme rainfall) during 24 h representing the rainy season (Wang et al., 2024). Three replicates for both rainfalls were carried out at each site. To decrease errors and uncertainties due to soil heterogeneity, two infiltration experiments were conducted simultaneously in plot number 5 for each forest type (Fig. 2 (c), (d)), and only one infiltration experiment was conducted in the 4 other plots (Fig. 2 (a), (b)). That is, 35 mm rainfall was infiltrated in plot number 1, 2, and 5-1, while 60 mm was infiltrated in plot number 3, 4, and 5-2.

The selected experimental sites were generally selected to be representative of the two different forest types with flat surface to avoid effects of tree trunks on the infiltration. We ensured that there were no rainfall events for three days before and after the infiltration. The surface litter of the plots was removed, and weeds were carefully removed. Subsequently, we carefully pushed stainless steel metal frames of 1.1 m × 0.7 m (plot 1–4) and 1.1 m × 1.2 m (plot 5) into the soil to a depth of about 10 cm. The dual-tracer was a mixture of 5 g L⁻¹ Brilliant Blue FCF and 5 g L⁻¹ NaBr solution. The rainfall was uniformly applied with a backpack electric sprayer at the rate of 40 mm/h to ensure that there

Table 1
Summary of experimental site information.

Site	Geographic position	Slope/%	Plant height/m	Average DBH/cm	Vegetation coverage/%	Dominant tree species	Groundcover
PF	41°18'14"N, 124°48'57"E	16–19	22.31	26.46	75	<i>Larix gmelinii</i> , <i>Pinus koraiensis</i>	<i>Leymus chinensis</i> , <i>Geranium wilfordii</i> , <i>Glycine soja</i> , etc.
NSF	41°18'18"N, 124°48'54"E	15–18	21.46	26.32	92	<i>Quercus mongolica</i> , <i>Acer pictum</i> , <i>Populus davidiana</i>	<i>Rubia chinensis</i> , <i>Urtica angustifolia</i> , <i>Viola philippina</i> , <i>Viola arcuata</i> , etc.

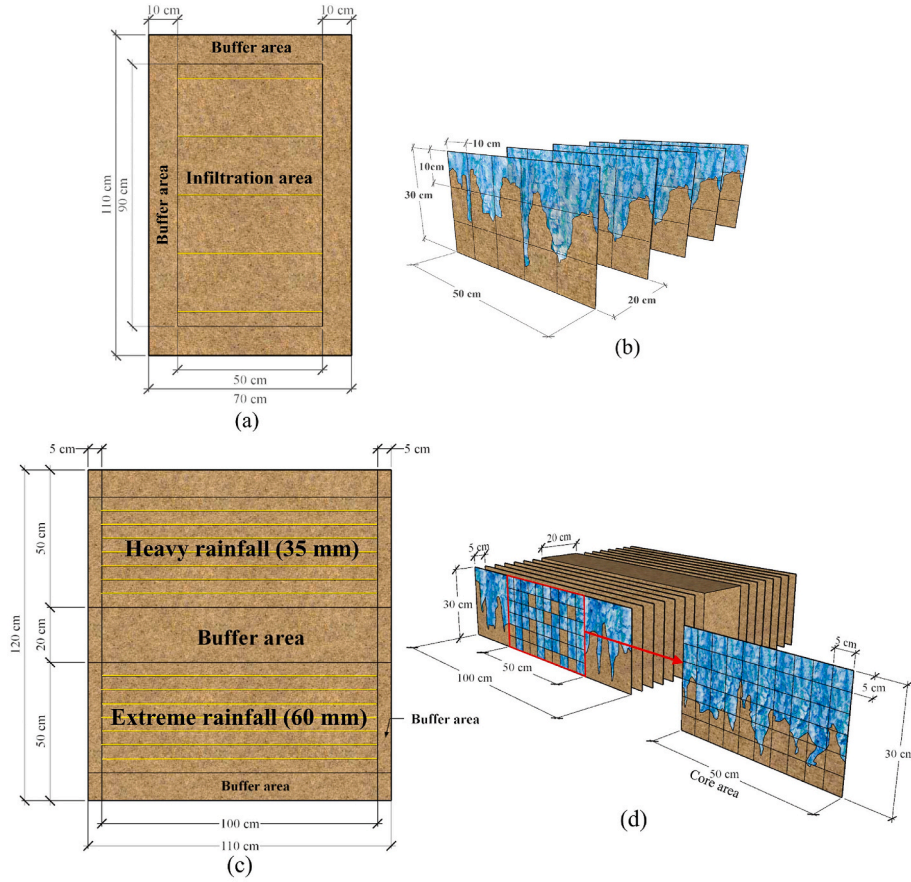


Fig. 2. Dual-tracer experimental set-up and soil solute sample collection layout. (a) and (c) top view of the dual-tracer experiments for plot 1–4 and plot 5-1 and 5-2 in the two different forest restoration areas, respectively. The yellow lines represent the location of the dug vertical soil profiles. (b) and (d) vertical soil profile diagram corresponding to (a) and (c), respectively. The smallest square cell (10 cm × 10 cm for plot 1–4 and 5 cm × 5 cm for plot 5-1 and 5-2) is shown for each plot type. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

was no ponding during the infiltration process. The subsequent uniformity coefficient was 0.95. The infiltration area was covered with a tarpaulin after spraying the tracers to prevent external precipitation and surface evaporation. The tarpaulin and metal frame were carefully removed after 24 h of the tracer experiment and vertical soil profiles of each plot were dug as shown in Fig. 2, with 20 cm intervals between the profiles for plot 1–4 and 5 cm intervals between the profiles of plot 5 to visualize solute water flow pathways. Consequently, 5 profiles were dug in each of the plot 1–4 and 7 profiles in plot 5-1 and 5-2 (a total of 68 profiles). Horizontal and vertical steel rulers were placed over each profile as marker and a digital camera (Canon 700D) was used to take photographs of the profile surfaces, with no less than 3 photos per profile.

2.3. Extraction of parameters

2.3.1. Preferential flow pathway parameters for soil profiles

The preferential flow parameters were extracted according to the stained soil profile images. By using the “perspective cutting tool”–“adjustment–brightness/contrast”–“color replacement tool” by

Photoshop (2020) software, dyed areas were substituted with black and undyed areas with white. Gravel parts protruding out from the profile were also replaced with white. Finally, grayscale images with a resolution of 1 pixel mm^{−1} were exported. Subsequently, the dyed profile binary data were obtained at pixel size by using Matlab 2020. Finally, the preferential flow morphology parameters (*DC*, *CV*, *PF-fr*, *UniFr*, and *LI*) were calculated.

Dye coverage (*DC*, %), which characterized the proportion of dyed area in the profile, was calculated (*D*, mm²) for each profile image (Flury et al., 1994):

$$DC(\%) = D(\text{mm}^2) / [D(\text{mm}^2) + ND(\text{mm}^2)] \times 100 \quad (1)$$

where *ND* (mm²) is unstained area. Coefficient of variation (*CV*, %) of *DC* is the degree of preferential flow where lower preferential flow is represented by a higher *CV*. Counted soil layer number (*n*), *DC* of layer *i* (*DC_i*, %) and average *DC* of the soil profile ($\overline{DC}$, %), *CV* (%) were calculated according to (Luo et al., 2022):

$$CV(\%) = \sqrt{\left[\frac{1}{(n-1)} \sum_{i=1}^n (DC_i(\%) - \overline{DC}(\%))^2 \right] / (n-1)} / 1 / n \sum_{i=1}^n DC_i(\%) \quad (2)$$

The matrix flow depth (*UniFr*, mm) represents the soil depth when *DC* is reduced to 80% along the soil profile. The area above this depth of soil was rearded as the matrix flow region (Van Schaik, 2009).

The preferential flow fraction (*PF-fr*, %), which characterizes the proportion of water flow area infiltrated by all preferential flow paths on the profile (Van Schaik, 2009), was calculated using *UniFr* (mm), soil vertical profile width *W* (mm), and total profile area *TSA* (mm²):

$$PF-fr(\%) = (1 - UniFr(mm) \bullet W(mm) / TSA(mm^2)) \times 100 \quad (3)$$

The length index (*LI*) represents the sum of the absolute values of the difference between *DC*_{*i+1*} (%) and *DC*_{*i*} (%) across two consecutive layers (1 mm interval in this study) on the vertical profile of the soil (Bargués Tobella et al., 2014):

$$LI(\%) = \sum_{i=1}^n |DC_{i+1}(\%) - DC_i(\%)| \quad (4)$$

2.3.2. Evaluation of spatial variability of preferential flow

Based on the profile data of plot 5-1 and 5-2, we obtained the plot-scale spatial morphological characteristics of preferential flow by the difference method and analyzed the spatial variability. We evaluated the spatial variability of preferential flow pathways by using semi-variograms (Luo et al., 2019), which is a commonly used model to describe the spatial variability of soil properties in geostatistics. The dyed images of each profile at each plot for NSF5-1, 5-2 and PF5-1, 5-2 were divided into 120 units using 5 cm × 5 cm, and *DC* data of each unit were extracted. The Kolmogorov-Smirnov test of IBM SPSS Statistics 22.0 software was used to test the normal distribution of the data (*p* < 0.05). The data sets that did not pass the test were normally transformed by Minitab 21.1 software (Minitab Inc., State College, PA, USA). The GS + v.9.0 software (Gamma Design Software LLC., Plainwell, MI, USA) was used to analyze the semi-variance for all data conforming to the normal distribution according to:

$$\gamma(l) = 1 / 2N(l) \sum_{i=1}^{N(l)} [A(x_i) - A(x_i + l)]^2 \quad (5)$$

where *N*(*l*) denotes the number of samples with sampling distance *l*; *A*(*x*_{*i*}) and *A*(*x*_{*i*} + *l*) represent *DC* values of the variable *A*(*x*) at *x*_{*i*} and *x*_{*i*} + *l*, respectively.

According to the principle of the highest coefficient of determination and the lowest residual error, we selected the spherical model, exponential model, and gaussian model as optimal models for different plots explaining the spatial variability of the preferential flow in the vertical and horizontal direction:

Spherical model:

$$\gamma(l) = \begin{cases} 0, l = 0 \\ C_0 + C(3l/2a - l^3/2a^3), 0 < l \leq a \\ C_0 + C, l > a \end{cases} \quad (6)$$

Exponential model:

$$\gamma(l) = \begin{cases} 0, l = 0 \\ C_0 + C(1 - e^{-l/a}), l > 0 \end{cases} \quad (7)$$

Gaussian model:

$$\gamma(l) = \begin{cases} 0, l = 0 \\ C_0 + C(1 - e^{-l^2/a^2}), l > 0 \end{cases} \quad (8)$$

where *C*₀ is nugget variance, which characterizes the spatial heterogeneity and measurement error of variables below the minimum sampling scale; *C*₀ + *C* is structural variance sill characterizing the overall degree of variation of the preferential flow pathways; *C*₀/*C*₀ + *C* represents the spatial dependence of preferential flow pathways, with a range of values divided into three classes: <25% (strong spatial dependence), 25–75% (middle spatial dependence), >75% (weak spatial dependence); *A* is

spatial autocorrelation range, i.e., the autocorrelation range of the regionalized variable; *R*² is coefficient of determination with a value closer to 1 indicating a better model fit; *RSS* is residual sum of squares, the smaller the value, the better the model adaptation.

2.4. Soil and root sampling measurements

The soil samples were collected by cutting cylinder (volume, 100 cm³) with 10 cm interval along the soil depth in 3 layers outside the metal frame corresponding to the 1st, 3rd, and 5th profile positions of plots 1, 2, and 5-1. Two samples were taken from each layer of each profile, one of which was used for roots collection, and the other for the determination of soil *BD*, total porosity (*TP*), and *ISWC*. The undisturbed soil samples were collected in situ by aluminum boxes at the same position, and the soil particle size composition was determined by Malvern Mastersizer-3000 (Malvern Instruments Ltd., Malvern, UK).

The weighed soil from the cutting cylinder with initial wet soil (*M*_{wet}, g) was returned to the laboratory to be dried in an oven at 105 °C for 24 h and again weighed (*M*_{dry}, g) (cylinder *M*_{comp} is mass of the empty cutting cylinder, g) to calculate the soil *BD*:

$$BD(g \cdot cm^{-3}) = \frac{M_{dry}(g) - M_{comp}(g)}{100(cm^3)} \quad (9)$$

Soil *TP* was calculated based on the assumption that soil pores were filled with water and no air, as evidenced by *BD* and particle density of 2.65 g cm⁻³ (Danielson and Sutherland, 1986):

$$TP(\%) = \left(1 - \frac{BD(g \cdot cm^{-3})}{2.65}\right) \times 100\% \quad (10)$$

In addition, the *ISWC* was calculated as:

$$IWC(\%) = \frac{M_{wet}(g) - M_{dry}(g)}{M_{dry}(g) - M_{comp}(g)} \times 100\% \quad (11)$$

Sample collection for soil solute concentration determination was carried out as shown in Fig. 2 (b), (d), by taking soil samples in the smallest square cell of the grid (10 cm × 10 cm in plot 1–4 and 5 cm × 5 cm in plot 5-1 and 5-2). Each sample corresponded to about 70 g and numbered sequentially.

The cutting cylinder used for collecting root samples was brought back to the laboratory, and each sample was rinsed with distilled water on a 0.25 mm sieve to carefully retain all roots. The roots were scanned by a root scanner (EPSON V850), and scanned images were analyzed by Win RHIZO 2020a software (Regent Instruments Inc., Quebec City, Canada). Root architecture indices were calculated to describe root characteristics, including root length density (*RLD*), root surface area density (*RSAD*), root volume density (*RVD*), and root mean diameter (*RMD*).

2.5. Determination of soil solute concentration

The collected soil samples for measuring soil solute concentration were dried at 105°, ground, and passed through a 1 mm sieve. 10 g of each soil sample were mixed with 50 mL of deionized water, shaken for 12 h to ensure homogeneity. The soil solution was then centrifuged at 3000 rpm and the supernatant was filtered. The obtained solution was measured by a bromine ion meter (PX5J-216 F, Shanghai Leici) to determine the concentration of Br⁻ (Luo et al., 2019). For better comparison of solute migration patterns at the different profiles and to avoid influence from soil spatial heterogeneity, we utilized the relative concentration (*C*^{*}, %) to represent the measured Br⁻ concentration (*C*, mg·kg⁻¹) (Schwen et al., 2014), and maximum and (*C*_{max}, mg·kg⁻¹) and minimum concentration (*C*_{min}, mg·kg⁻¹) on the profile grid were calculated according to:

$$C'(\%) = \frac{C(\text{mg} \cdot \text{kg}^{-1}) - C_{\min}(\text{mg} \cdot \text{kg}^{-1})}{C_{\max}(\text{mg} \cdot \text{kg}^{-1}) - C_{\min}(\text{mg} \cdot \text{kg}^{-1})} \times 100\% \quad (12)$$

2.6. Preferential flow pathway parameters

The diversity parameters of the tracer relative solution concentration distribution were used to represent the characteristics of preferential flow pathways for each profile according to:

Simpson index (D):

$$D = 1 - \sum_{i=1}^S p_i^2 \quad (13)$$

Shannon-Wiener index (H):

$$H = - \sum_{i=1}^S p_i \ln p_i \quad (14)$$

Margalef index (F):

$$F = (S - 1) / \ln N \quad (15)$$

Pielou index (J):

$$J = H / \ln S \quad (16)$$

where S is the total number of relative solution concentrations in the profile total grid, p_i is the proportion of the type i relative solution concentration to the total number of relative solution concentration in the total grid, and N is the total number of relative solution concentration in the total grid.

2.7. Multi-indicators of preferential flow development

A comprehensive evaluation indicator of preferential flow development (preferential flow index, PFI) was designed by combining preferential flow morphological parameters and diversity parameters (solute migration characteristic parameters). The normalized dimensionless value Z_{ij} of preferential flow pathway morphological parameters (DC , CV , $Unifr$, $PF-fr$, and Li) and diversity parameters (D , H , F , and J) of different forest restoration forestlands (PF and NSF) were calculated by the range method. The index weight W_j was determined by mean squared error decision. Finally, PFI was calculated. The calculation were done as follows.

(1) Parameter standardization (normalization):

$$Z_{ij} = (X_{ij} - X_{\min}) / (X_{\max} - X_{\min}) \quad (17)$$

$$Z_{ij} = (X_{\max} - X_{ij}) / (X_{\max} - X_{\min}) \quad (18)$$

where Eqn. (17) was used for positive indicators and Eqn. (18) for negative indicators. X_{ij} is the measured preferential flow pathway indicators; $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the preferential flow pathways indicators, respectively; i is the serial number of n data of the preferential flow pathway indicators; and j is the number of indicator ($j = 1, 2, 3, \dots, 8$).

(2) Mean value of standardized indicator:

$$E(G_j) = 1 / n \sum_{i=1}^n Z_{ij} \quad (19)$$

(3) Indicator G_j mean square error:

$$\sigma(G_j) = \sqrt{1 / n \sum_{i=1}^n (Z_{ij} - E(G_j))^2} \quad (20)$$

(4) Indicator G_j weight coefficient:

$$W_j = \sigma(G_j) / \sum_{j=1}^8 \sigma(G_j) \quad (21)$$

(5) Comprehensive evaluation index of preferential flow development degree (PFI):

$$PFI = \sum_{i=1}^n Z_{ij} W_j \quad (22)$$

2.8. Data analysis

The normality of each indicator was tested by using Shapiro-Wilk test, and the non-normal distribution data were transformed by logarithm or square root. One-way ANOVA was used to assess the effects of different forest restoration approaches and soil layers on soil physical properties, root traits, and solute migration in the profile ($p < 0.05$). Preferential flow pathway diversity parameters were calculated by Past 4.09 software. The independent-sample t -test was used to compare the differences in preferential flow pathway morphology and path diversity parameters between infiltration amounts and forest restoration approaches ($p < 0.05$). Pearson correlation analysis was used to clarify the effects of soil physical properties and root traits on the preferential flow development. In addition, we further explored the direct and indirect effects of soil and vegetation on the preferential flow development through path analysis. All above statistical analyses were done through IBM SPSS Statistics 22 (SPSS Inc., Chicago, Illinois, USA) software, and the relevant plotting was performed by Origin (2023b) (OriginLab, Northampton, Massachusetts, USA) software.

3. Results

3.1. Soil physical properties and root traits

The soil physical properties of forestland under the two forest restoration approaches exhibited specific differences. The BD of PF exhibited a significant increase (6.8%) compared to NSF, yet the difference was not statistically significant ($p > 0.05$). Meanwhile, the TP of PF was significantly lower than that of NSF ($p < 0.05$), specifically indicating a decrease of 2.4% compared to NSF. The $ISWC$ between the two sites was relatively similar, showing no significant difference ($p > 0.05$), with PF slightly lower by 0.2% compared to NSF. The soil particle size composition content demonstrated significant disparities at both sites ($p < 0.05$). The sand content was higher in NSF (70.9%), while PF exhibited higher silt and clay contents by 2.2% and 1.4%, respectively, compared to NSF (Table 2).

Divergent root traits were observed at the different forest sites (Fig. 3). The RLD and $RSAD$ for each site exhibited a pronounced decrease with soil depth, particularly in the 10–20 cm soil layer, where RLD decreased by 68.6% and 49.1% for PF and NSF, respectively, and $RSAD$ decreased by 54.1% and 32.6%, respectively (Fig. 3 (a), (b)).

Table 2

Soil physical properties at the 0–30 cm soil depth in different forest restoration sites.

Site	$BD/\text{g} \cdot \text{cm}^{-3}$	$TP/\%$	$ISWC/\%$	Particle size composition		
				$Sand/\%$	$Silt/\%$	$Clay/\%$
PF	$1.42 \pm 0.11a$	$51.75 \pm 0.94a$	$25.56 \pm 1.17a$	$67.33 \pm 2.33a$	$22.37 \pm 1.94b$	$10.30 \pm 0.81b$
	$1.33 \pm 0.09a$	$53.02 \pm 0.09b$	$25.61 \pm 0.89a$	$70.87 \pm 2.62b$	$20.17 \pm 2.33a$	$8.95 \pm 0.61a$

Note: Values are means $\pm$ SD ($n = 9$). BD , bulk density; TP , total porosity; $ISWC$, initial soil water content; $Clay$, clay content; $Silt$, silt content; $Sand$, sand content. Different letters a and b indicate significant differences ($p < 0.05$) between different forest restoration sites.

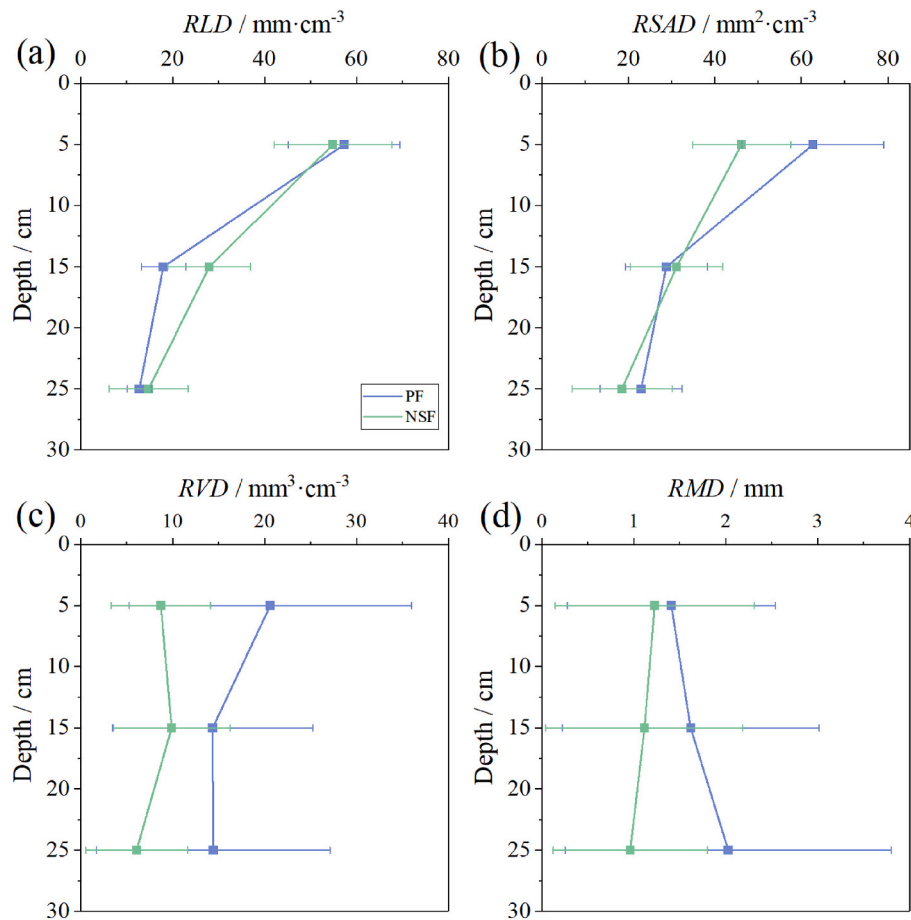


Fig. 3. Changes in the root traits with soil depth in the different forest restoration sites. *RLD*, root length density; *RSAD*, root surface area density; *RVD*, root volume density; *RMD*, root mean diameter.

Significant differences were noted in *RLD* and *RSAD* among different soil layers within the same site ($p < 0.05$). In terms of *RLD*, the overall trend indicated lower values at PF as compared to NSF, particularly in the 10–20 cm layer, where statistical differences between the two sites were observed ($p < 0.05$) (Fig. 3 (a)). For *RSAD*, a general trend showed lower values at NSF compared to PF. In the surface soil layer (0–10 cm), NSF was significantly lower than PF ($p < 0.05$), while along the soil profile, differences between the two sites diminished ($p > 0.05$) (Fig. 3 (b)). At soil depth of 0–30 cm, the *RVD* exhibited an overall significant lower value at NSF compared to PF ($p < 0.05$), particularly in the surface layer (0–10 cm) ($p < 0.05$). However, variations in *RVD* along the soil profile within the same site were not significant ($p > 0.05$) (Fig. 3 (c)). Regarding *RMD*, the overall trend indicated slightly lower at NSF compared to PF. Nevertheless, there were no significant differences between the two sites or among different soil layers within the same site ($p > 0.05$). The trend in *RMD* variation was slightly different between the two sites, with PF showing an increasing trend in *RMD* along the soil profile, while NSF exhibited the opposite trend (Fig. 3 (d)).

3.2. Morphological characteristics of preferential flow pathways

3.2.1. Preferential flow pathway morphological characteristics

We selected a typical vertical profile from each site for presentation (Fig. 4 (a)–(l)), both sites had a large horizontal spread of dye tracer within the surface 5 cm of the soil, demonstrating a strong interaction between matrix flow and preferential flow within this soil layer. Combined with dye coverage ratio (*DC*) (Fig. 5 (a)–(l)), both sites started to diverge sharply from about 5 cm depth and began to develop significant preferential flow phenomenon. Besides, the two forestland preferential

flow patterns exhibited differences in morphology, with the PF showing a ‘funnel’ shape with fewer and wider flow pathways, and the NSF showing a ‘finger’ shape with more and narrower flow pathways. As infiltration increased, both vertical connectivity and horizontal extensibility of the preferential flow pathways increased. Notably, preferential flow pathways that isolated rather than developed from the surface occurred at each site.

We further analyzed the effect of different forest restoration type for different infiltration amount on the preferential flow parameters (Fig. 5 (m)–(q)) to better quantify the morphological development of preferential flow pathways. Generally, *DC* showed a nonlinear decreasing trend with increasing soil depth across the sampled sites, but the trends were slightly different, mainly due to the difference of water infiltration characteristics caused by the strong spatial heterogeneity of soil structure. The mean *DC* values of the NSF sites were 13.9% and 10.1% higher than those of the PF sites under 35 mm and 60 mm infiltration conditions, respectively, but there was no significant difference in *DC* between the two forest restoration types ($p > 0.05$). Increasing infiltration from heavy to extreme rainfall significantly increased *DC* ($p < 0.01$) for both forest types (Fig. 5 (a)–(m)). The mean value of *CV*, which was used to characterize the dye difference between preferred flow and matrix flow, showed that PF (0.59 ± 0.15) was approximately equal to NSF (0.59 ± 0.20) for 35 mm infiltration, and PF (0.35 ± 0.14) was slightly greater than NSF (0.29 ± 0.12) for 60 mm infiltration. However, under different infiltration conditions, the mean *CV* of the same site had remarkably significant differences ($p < 0.01$) (Fig. 5 (n)). The average depth of *UniFr* in the PF was 47.9 ± 37.6 cm for 35 mm infiltration, which was lower than NSF (65.5 ± 28.0 cm). Note that when infiltration increased to 60 mm, the *UniFr* of PF (76.1 ± 31.3 cm) was significantly lower than that

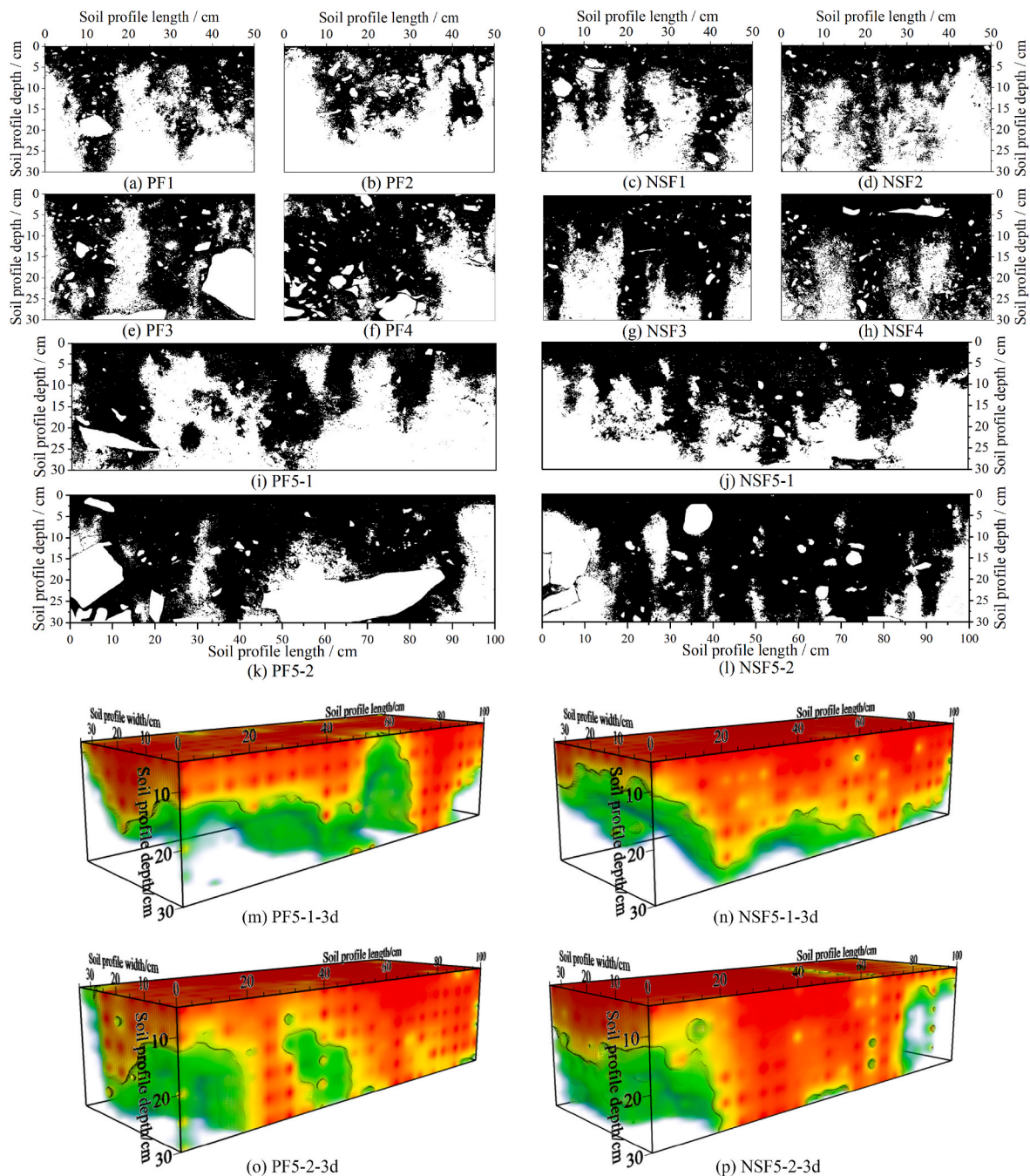


Fig. 4. Morphological characteristics of preferential flow pathways in different forest restoration plots. (a)–(l) representative dyed soil profile images of each site. PF1, 2, 5-1 and NSF1, 2, 5-1, plantation forest or natural secondary forest under 35 mm infiltration amount, respectively. PF3, 4, 5-2 and NSF3, 4, 5-2, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively. (m)–(p) spatial morphological characteristics of preferential flow pathways under different forest restoration approach and infiltration amounts.

of NSF (111.4 ± 47.8 cm) ($p < 0.05$). This indicates that preferential flow begins earlier in PF regardless of infiltration conditions. Furthermore, increasing infiltration can significantly increase the *UniFr* for the same forest type ($p < 0.05$) (Fig. 5 (o)). The mean value of *PF-fr* showed that 78.2, 74.6 and 62.9% of the dyed area were corresponding to the preferential flow pathways under infiltration of 35 mm at the NSF plot, and 60 mm at the PF and NSF plots, respectively. This value increased to 84.0% under the 35 mm infiltration for the PF plot. In both types of forests, increasing infiltration significantly reduced *PF-fr* ($p < 0.05$), and as infiltration increased up to extreme rainfall (60 mm), PF had a significantly higher value of *PF-fr* than NSF ($p < 0.05$) (Fig. 5 (p)). This indicates that a greater proportion of infiltrated water at PF stands is

transported through preferential flow channels. *LI*, the indicator used to quantify the heterogeneity of dye morphology due to preferential flow, was higher for PF (337% and 352%) than for NSF (341% and 310%), and there was no significant difference in *LI* under different infiltration conditions ($p > 0.05$) (Fig. 5 (q)).

3.2.2. Preferential flow pathway morphology

The presence of preferential flow influences the redistribution of soil moisture and solutes. To better understand the spatial variability of preferential flow pathways in the study area, we investigated the spatial distribution characteristics of preferential flow for the two types of restored forest land under different infiltration amount, both

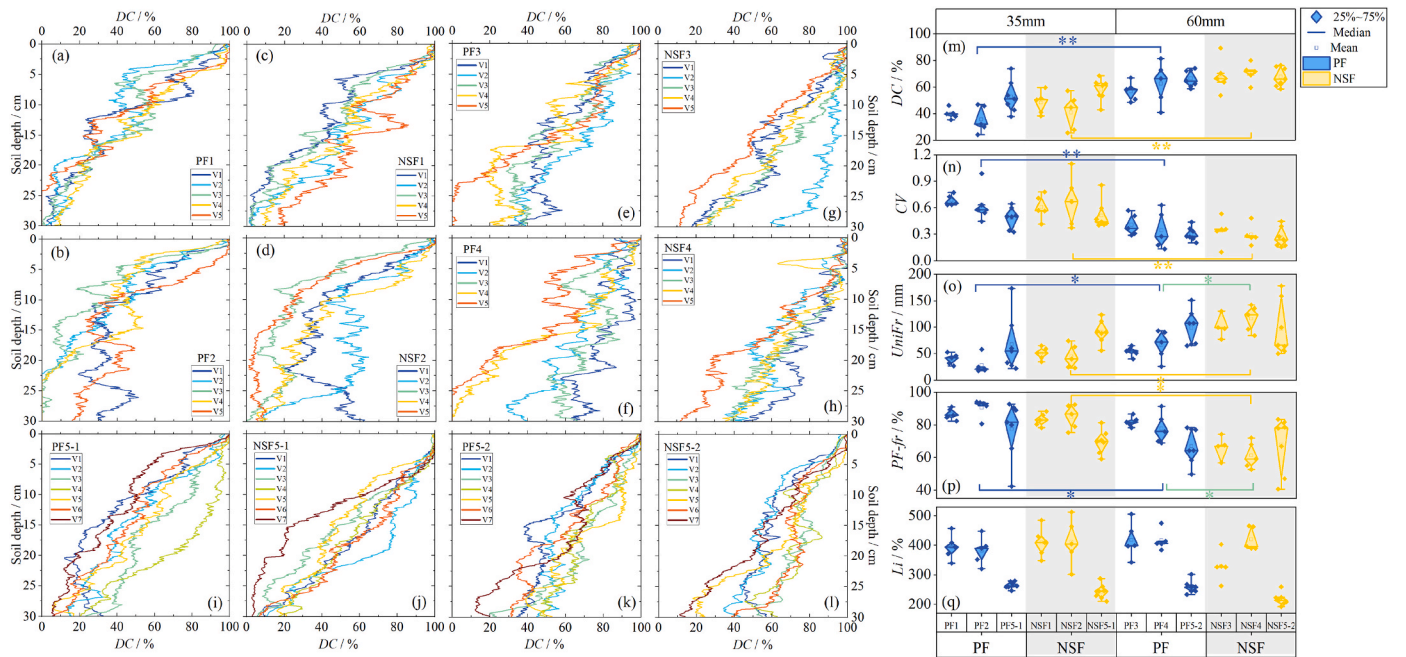


Fig. 5. Morphological characteristics of preferential flow pathways in different forest restoration plots. (a)–(l), trend of DC variation with increasing soil depth. PF1, 2, 5-1 and NSF1, 2, 5-1, plantation forest or natural secondary forest under 35 mm infiltration amount, respectively. PF3, 4, 5-2 and NSF3, 4, 5-2, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively. (m)–(q), box plots for DC, CV, UniFr, PF-fr, LI, respectively, in each site. Boxes with * indicate significant differences at $p < 0.05$, ** indicate significant differences at $p < 0.01$.

horizontally and vertically (Fig. 6). The isolines in Fig. 6 reflect the infiltration in the soil profile. In the horizontal direction, both PF and NSF preferential flows showed significant color block aggregation (Fig. 6 (a)–(d)), indicating that preferential flow pathways in different vegetation restoration forest land were characterized by aggregation and encircling. In the vertical direction, distinct “funnel” shaped color bands can be observed at all sites (Fig. 6 (e)–(h)). In addition, increasing infiltration amount could increase the horizontal and vertical connectivity of preferential flow pathways.

Furthermore, we analyzed the spatial heterogeneity by semi-variograms and fitting parameters (Table 3). The fitting effect was good ($0.903 \leq R^2 \leq 0.971$) for all models (spherical, exponential, and gaussian models). The spatial autocorrelation range (A) for NSF was larger than those of PF, both horizontally and vertically, indicating that the spatial variability of NSF preferential flow was stronger, especially in the horizontal direction. The results of $C_0/(C_0+C)$, which characterizes the proportion of total variation due to spatial heterogeneity influenced by random factors, showed that $C_0/(C_0+C)$ was less than 25% in both horizontal and vertical directions for all sites except NSF5-1-H, indicating strong spatial dependence of the variables, while NSF5-1-H was middle.

3.3. Solute migration characteristics of preferential flow pathways

3.3.1. Solute concentration distribution characteristics

The migration of Br^- in the soil profiles is shown in Fig. 7 and Table 4 with decreasing Br^- concentration with soil depth. Under 35 mm infiltration, Br^- concentration for PF was 1046 ± 383 , 590 ± 181 , and $397 \pm 222 \text{ mg kg}^{-1}$ at 0–10, 10–20, and 20–30 cm soil depths, respectively. For NSF, Br^- concentration was 1658 ± 747 , 604 ± 242 , and $439 \pm 195 \text{ mg kg}^{-1}$ at the same respective depths. Under 60 mm infiltration conditions, the trend of Br^- concentration change along soil profiles was consistent with that of 35 mm, with Br^- concentration in the 0–10 cm soil layer being 2.18 and 3.55 times higher than that in the 10–20 and 20–30 cm soil layers for PF, respectively. Furthermore, for NSF, Br^- concentration in the 0–10 cm layer was 2.2 and 3.7 times higher than

that in the 10–20 and 20–30 cm soil layer, respectively. Significant differences in solute concentration at different depths were observed between different forest restoration land under both 35 mm and 60 mm infiltration conditions ($p < 0.05$).

Comparing the solute migration of the two forest types, we found that the mean solute concentration of all soil layers for NSF was higher than that for PF under different infiltration conditions. Under the condition of heavy rainfall (35 mm), solute concentration for NSF was significantly higher than that for PF at 0–10 cm layer ($p < 0.05$), but there was no significant difference in Br^- concentration between the two forest types at 10–30 cm depth ($p > 0.05$). When increasing to storm rainfall intensity (60 mm), Br^- concentration in the 0–20 cm soil layer demonstrated significant differences between forest type ($p < 0.05$), whereas in the 20–30 cm layer, no significant difference was observed ($p > 0.05$). When comparing Figs. 7 and 8 for deeper soil (20–30 cm), it can be seen that partial profiles appear with DC equal to 0. However, the Br^- concentration was not always 0 in the corresponding unit grid region. This implies that solutes Br^- diffuse through the soil over a wider range than the observed Brilliant Blue staining pathways.

3.3.2. Preferential flow pathway characteristics

To better characterize the development of preferential flows, we used variation indices to evaluate solute distribution in the soil profiles. Considering that solutes are also transported along the preferential flow pathways, the solute concentration distribution variation parameters were employed to quantify the diversity of preferential flow pathways. As illustrated in Fig. 9, the four preferential flow pathways variation parameters for each plot were calculated based on Br^- relative concentrations. The mean of preferential flow variation parameters H , D , and J observed for the two forest types under varying infiltration amount showed that PF35 significantly exceeded the other plots ($p < 0.05$). Specifically, the mean H value of PF35 was 1.055, 1.048, and 1.049 times higher than NSF35, PF60, and NSF60, respectively; the mean D of PF35 was 0.021, 0.021, and 0.017 higher than the other three plots, respectively; and the mean J value of PF35 was 4.79%, 4.64%, and 4.91% higher than the other three plots, respectively. The F mean value

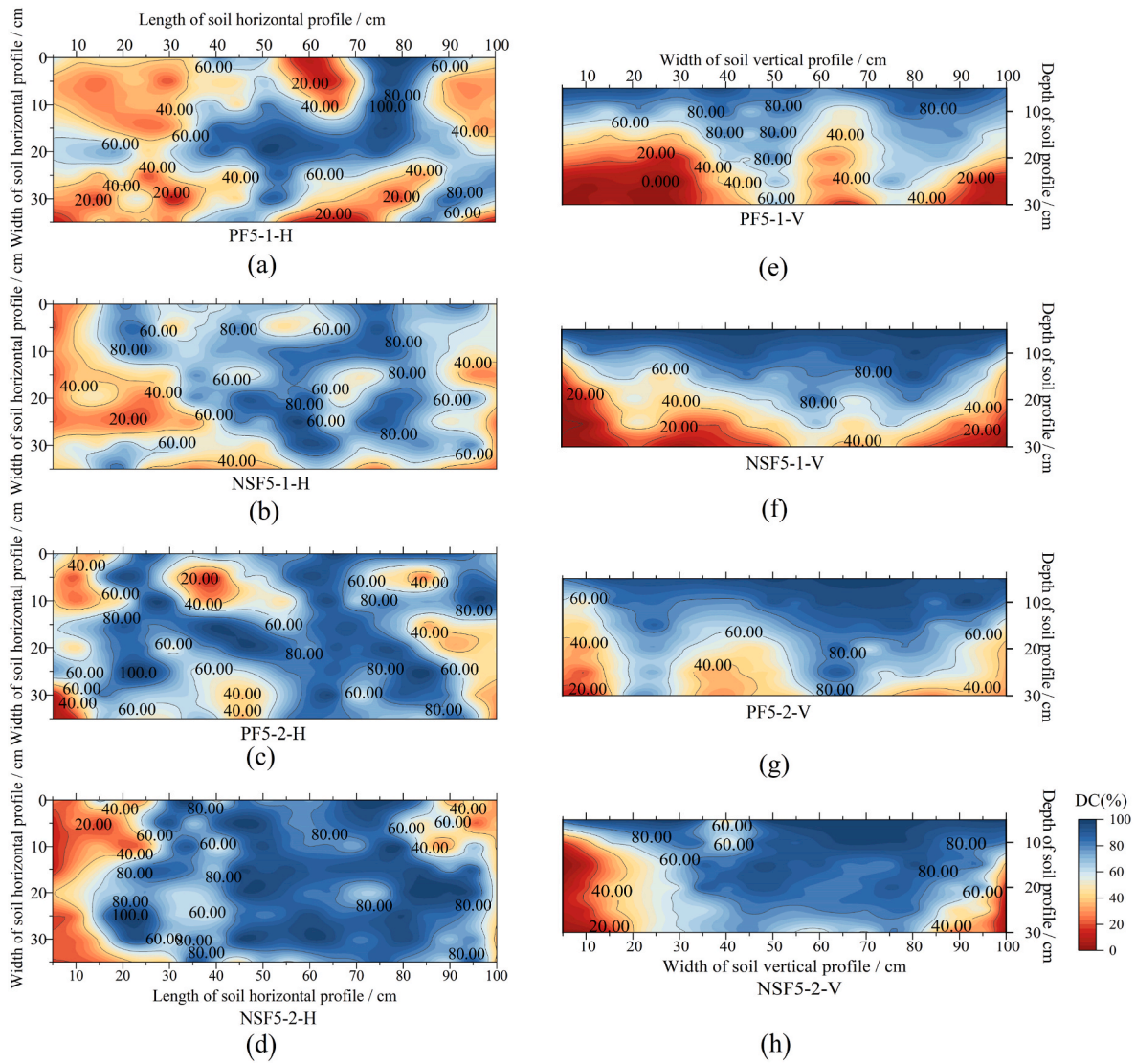


Fig. 6. Spatial distribution of preferential flow pathways in horizontal and vertical directions for different forest restorations and different infiltration amount. (a)–(d), NSF5-1-H, PF5-1-H and NSF5-2-H, PF5-2-H are the horizontal spatial distribution of dye coverage for NSF and PF under 35 mm and 60 mm infiltration amount, respectively; (e)–(h), NSF5-1-V, PF5-1-V and NSF5-2-V, PF5-2-V are the vertical spatial distribution of dye coverage in NSF and PF under 35 mm and 60 mm infiltration amount, respectively.

Table 3

Semi-variogram parameters showing the degree of spatial variability of preferential flow pathways under different forest restoration approach and different infiltration amount.

Land spatial type	Variogram model type	A (m)	Nugget (C_0)	Sill (C_0+C)	$C_0/(C_0+C)$ %	Spatial class	R^2	RSS
PF5-1-H	Sp	0.219	0.063	0.674	9.347	S	0.968	6.624E-03
NSF5-1-H	Ex	0.937	0.323	0.864	37.384	M	0.971	3.578E-03
PF5-2-H	Ex	0.174	0.019	0.520	3.654	S	0.927	3.664E-03
NSF5-2-H	Sp	1.188	0.395	1.655	23.867	S	0.906	4.520E-02
PF5-1-V	Ga	0.211	0.014	1.051	1.332	S	0.903	1.070E-01
NSF5-1-V	Ga	0.265	0.015	1.433	1.046	S	0.936	1.510E-01
PF5-2-V	Ga	0.253	0.037	0.824	4.490	S	0.919	5.820E-02
NSF5-2-V	Ga	0.310	0.182	1.301	13.989	S	0.955	6.950E-02

Note: Sp, spherical model; Ex, exponential model; Ga, gaussian model; A, spatial autocorrelation range; C_0 , nugget variance; (C_0+C), structural variance sill; S, strong spatial dependence; M, middle spatial dependence; R^2 , coefficient of determination; RSS, residual sum of squares; NSF5-1-H, PF5-1-H and NSF5-2-H, PF5-2-H are the horizontal spatial distribution of dye coverage in NSF and PF under 35 mm and 60 mm infiltration, respectively; NSF5-1-V, PF5-1-V and NSF5-2-V, PF5-2-V are the vertical spatial distribution of dye coverage in NSF and PF under 35 mm and 60 mm infiltration, respectively.

was the lowest for PF35 (2.00 ± 0.05) and there was no significant difference among the plots ($p > 0.05$). This Indicates that the PF plot under 35 mm infiltration was the most important in preferential flow.

The NSF plot under the same infiltration (35 mm) was important regarding preferential flow as compared to PF and had a similar diversity of preferential flow pathways as the NSF with 60 mm infiltration.

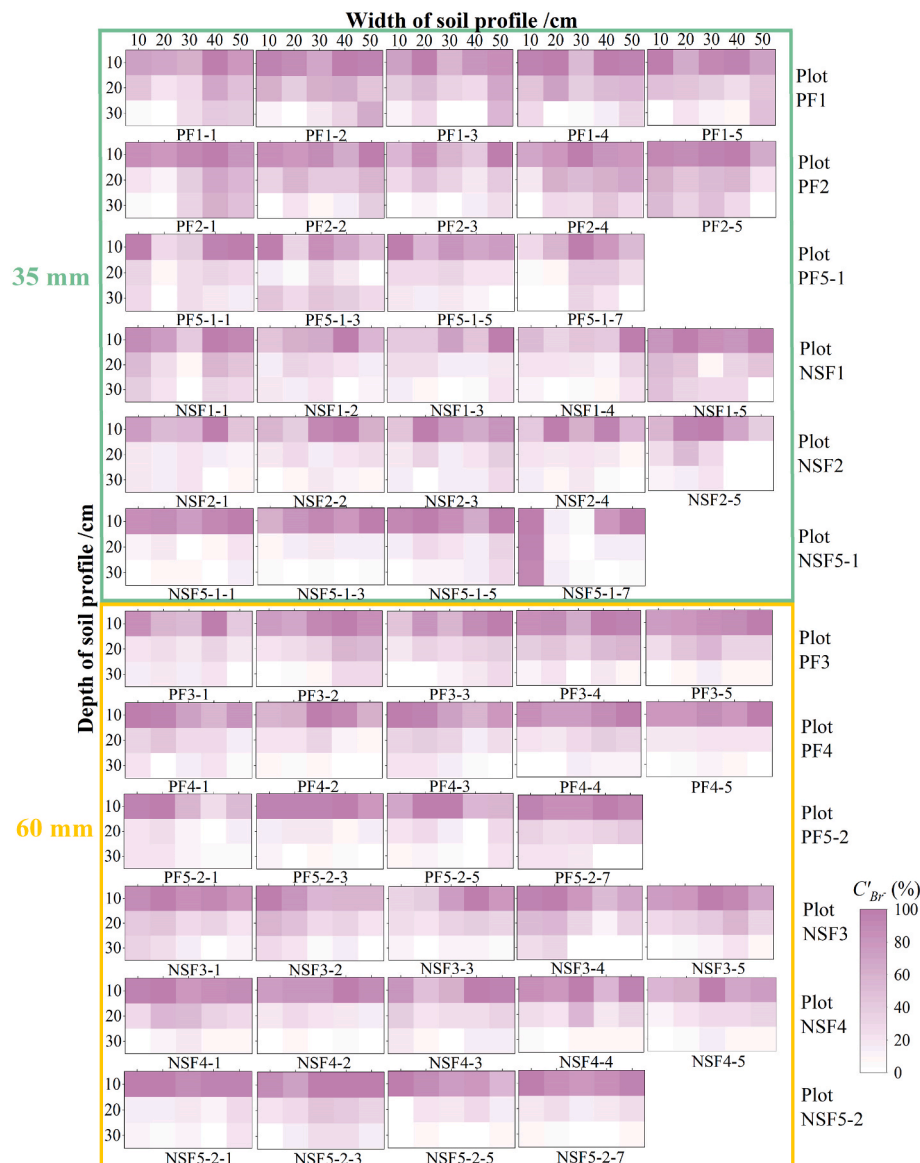


Fig. 7. Distribution characteristics of solute relative concentration for two different forest restoration plot soil profiles. PF1, 2, 5-1 and NSF1, 2, 5-1, plantation forest and natural secondary forest under 35 mm infiltration amount, respectively. PF3, 4, 5-2 and NSF3, 4, 5-2, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively. The last digit represents the number of each profile in the plot.

That is, changing the rainfall intensity had no marked effect on the preferential flow at the NSF site. In contrast, for PF sites, increasing infiltration can significantly change the preferential flow pathways and reduce the variation of the preferential flow path.

3.4. Evaluation of preferential flow development

Considering that preferential flow occurs as a process of rapid migration of water carrying solutes to deeper soil layers circumventing the majority of the soil matrix. However, we also noted in section 3.3 that the extent of solute diffusion was greater than the profile preferential flow staining pathways. Therefore, when assessing preferential flow development, not only the flow morphology of the soil profile should be considered, but also the solute distribution characteristics. Thus, when designing the *PFI*, we selected the preferential flow morphology parameters (including *DC*, *CV*, *UniFr*, *PF-fr*, and *Li*) and the preferential flow pathway variation parameters (including *H*, *D*, *F*, and *J*) to calculate the *PFI* for each plot through the multi-indicator method, and the weight coefficient of each indicator is shown in Table 5.

The mean *PFI* of PF35, NSF35, PF60 and NSF60 were 0.682, 0.567, 0.624, and 0.572 (Fig. 10), which were consistent with the preferential flow profiles and spatial patterns (Figs. 4–6). Among them, the preferential flow development in PF35 was significantly higher than for the NSF ($p < 0.05$), and the mean *PFI* of PF35 was 20.2%, 9.2%, and 19.1% higher than that of NSF35, PF60, and NSF60, respectively. The preferential flow development decreased significantly as the infiltration amount increased from 35 mm to 60 mm for PF, which indicated that there were significant differences in infiltration pathways under different infiltration conditions for PF. In contrast, for NSF, increasing infiltration amount barely changed the preferential flow development, implying that when infiltration resulted from the heavy rainfall (35 mm), the infiltration pattern tended to stabilize within the 0–30 cm soil depth.

3.5. Factors influencing preferential flow development

The influence of soil physical characteristics and root traits on the degree of preferential flow development can be evaluated by correlating

Table 4
Statistical characteristics of soil profile solute concentration in each plot.

Depth of soil profile (cm)	Plot	Br ⁻ (mg·kg ⁻¹)					Plot	Br ⁻ (mg·kg ⁻¹)				
		Max	Min	Mean	SD	CV		Max	Min	Mean	SD	CV
0–10	PF1	1132.86	623.13	900.284	138.02	0.15	NSF1	2115.50	762.55	1093.88	349.18	0.32
10–20		838.28	238.41	568.25	159.43	0.28		921.62	124.81	516.44	174.11	0.34
20–30		737.75	0	335.97	194.14	0.58		730.75	85.49	377.18	151.28	0.40
0–10	PF2	1111.25	565.30	854.67	139.40	0.16	NSF2	2503.41	549.45	1444.65	463.15	0.32
10–20		791.87	280.28	521.56	138.38	0.27		854.22	0	542.07	249.37	0.46
20–30		737.03	0	327.55	176.23	0.54		666.134	0	417.74	188.51	0.45
0–10	PF 5-1	2085.68	573.97	1468.44	470.38	0.32	NSF 5-1	3199.74	1785.36	2628.99	359.16	0.14
10–20		1026.47	203.78	701.55	206.37	0.29		1166.15	406.51	789.43	209.31	0.27
20–30		916.42	147.91	560.18	231.15	0.41		925.02	113.14	542.26	216.55	0.40
0–10	PF3	2591.89	1271.77	2015.36	382.35	0.19	NSF3	3010.67	1052.60	1993.66	540.52	0.27
20–30		1608.23	712.75	1078.31	250.15	0.25		1926.17	661.61	1170.07	285.45	0.24
20–30		922.13	216.80	581.86	216.61	0.36		1176.08	386.67	708.34	233.49	0.33
0–10	PF4	2667.01	1642.19	2107.39	345.80	0.16	NSF4	3298.78	1638.07	2618.48	496.57	0.19
10–20		1366.43	539.88	908.27	227.85	0.25		1979.68	558.40	1173.95	342.39	0.29
20–30		908.14	325.14	551.32	195.75	0.36		1011.96	214.74	584.72	178.46	0.31
0–10	PF 5-2	3017.36	1013.35	2113.98	504.77	0.24	NSF 5-2	3800.13	2189.76	2858.33	401.54	0.14
10–20		1121.75	350.67	852.59	213.72	0.25		1637.55	466.85	1077.57	345.94	0.32
20–30		977.36	0	633.48	279.61	0.44		1208.66	101.89	713.96	337.25	0.47

Note: PF1, 2, 5-1 and NSF1, 2, 5-1, plantation forest or natural secondary forest under 35 mm infiltration amount, respectively. PF3, 4, 5-2 and NSF3, 4, 5-2, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively. Max, maximum value; Min, minimum value; Mean, mean value; SD, standard deviation; CV, coefficient variation.

PFI with environmental variables related to the soil rhizosphere layer. The Pearson correlation demonstrated that *PFI* was influenced by a variety of soil physical characteristics and root traits factors. Specifically, *PFI* had significantly positive correlation with clay content ($p < 0.001$), *RMD*, *RVD* ($p < 0.01$), and *TP* ($p < 0.05$), and significantly negative correlation with sand content ($p < 0.01$). Moreover, *ISWC*, silt content, *BD*, *RSAD*, *RLD* had no significant correlation with *PFI* ($p > 0.05$) (Fig. 11). Further path analysis of the relevant parameters showed that the degree of preferential flow was mainly influenced by clay content, *TP*, and *RVD* (Table 6). In general, clay content, *TP*, and *RVD* had positive effects on *PFI* with sum coefficients of 0.824, 0.462, and 0.624, respectively. Among them, clay content, *TP*, and *RVD* had direct positive effects on *PFI*, with direct path coefficients of 0.463, 0.487, and 0.378, respectively. Clay content and *RVD* had an indirect positive effect on *PFI* with path coefficients of 0.361 and 0.246, respectively. However, *TP* has an indirect negative effect on *PFI* with a path coefficient of -0.025 . In addition, the direct effect of clay content, *TP*, and *RVD* on *PFI* was greater than the indirect effect.

4. Discussion

4.1. Difference in soil and root properties for different forest restoration approach

Soil physical properties are broadly considered affected by numerous factors such as land use, forest restoration, and site conditions, which in turn exhibit strong spatial heterogeneity (Yimer et al., 2008; Sun et al. 2018a). Moreover, soil physical properties are also important variables that restrict the utilization of water and nutrients by vegetation, water and solute transport, and groundwater recharge (Zhang et al., 2022). Our study revealed that there were differences in soil physical properties for different forest restoration approaches, that is, the establishment of PF significantly altered the soil physical properties of forestland (Table 2), especially *TP* and particle size composition ($p < 0.05$). Due to excessive anthropogenic interventions such as frequent trampling, PF will lead to deformation of the soil surface layer due to increased surface contact pressure, which in turn causes an increase in *BD* as well as a decrease in *TP* (Alaoui et al., 2011b). At the same time, it will also change the soil structure, the connectivity of macropores and the resistance of root penetration (Beven and Germann, 2013; Sun et al., 2018b), which is consistent with the conclusion of our study. In addition, in shallow layer soil, soil fauna and microbial species and activities

are different in the two forestlands, and differences in root traits are key factors causing differences in soil *TP* (Chen et al., 2018). The differences in soil particle size composition between the different forest restoration approaches might be due to differences in soil erosion in the forestland induced by vegetation cover, climatic factors (wind, rainfall, and other external forcings) (Teague et al., 2011; Du et al., 2019). Although the two sites did not show a statistical difference ($p > 0.05$) in terms of *BD*, we found that *BD* of PF was higher than that of NSF, suggesting that artificial restoration approach is able to produce rapid and stable improvement in soil structure, which, in addition to the role of anthropogenic external disturbances, is also related to biological and physical processes such as root development, microbial and soil aggregate production in the soil (Phalempin et al., 2021).

Significant effects of different forest restoration approaches on the soil surface layer also include differences in the root traits of different vegetation. Root distribution can change the shape, size, and connectivity of the soil *TP*, which has a definite effect on soil infiltration (Chandler and Chappell, 2008). According to our results, PF vegetation roots had smaller *RLD*, larger *RSAD*, *RVD*, and *RMD*. In the 0–30 cm soil layer, PF vegetation roots were thicker and had a larger spatial proportion than for NSF, that is, PF root macropore volume was higher than that for NSF. The *RMD* of the two forestlands showed an opposite trend along the soil profile, with NSF showing a weak decreasing trend, which is consistent with the conclusion of previous studies that broad-leaved vegetation roots are enriched in shallow soil (0–25 cm) while coniferous roots have a larger downward extension (Bauhus and Messier, 1999). In addition, in conjunction with the field investigation in the study area we found that soil erosion occurs to varying degrees in different forestlands, but especially in the NSF plots. Due to the occurrence of soil erosion, the surface soil exposed some vegetation roots due to rainfall (hydraulic erosion), resulting in the root distribution converging in the surface layer, especially the thicker roots, which is another reason for the high *RMD* in the surface layer. In contrast, the soil and water conservation function of PF plays a certain role, there were no obvious bare vegetation coarse roots on the surface, and the root diameter tended to increase with soil depth. Evidently, the establishment of plantation forests in the study area performed a relatively good role on soil erosion control, and the vegetation showed better stability, which is concordant with the findings of Hua et al. (2022) on the lower risk of soil erosion in plantation forests. Consequently, in the study area, the management of NSF should appropriately increase prevention measures of soil erosion to ensure the stability of the rhizosphere soil

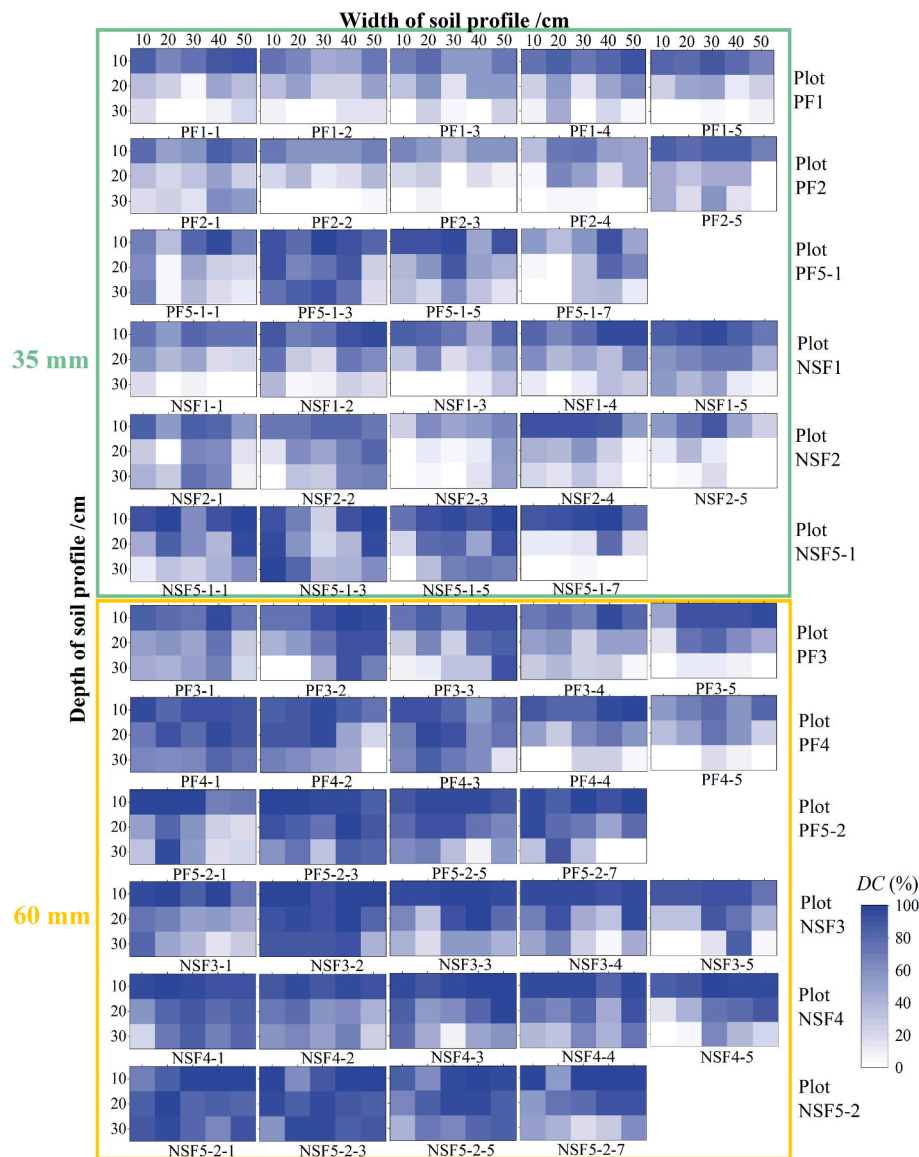


Fig. 8. DC distribution characteristics corresponding to relative solute concentration (Fig. 7) in each profile. PF1, 2, 5-1 and NSF1, 2, 5-1, plantation forest and natural secondary forest under 35 mm infiltration amount, respectively. PF3, 4, 5-2 and NSF3, 4, 5-2, plantation forest and natural secondary forest under 60 mm infiltration amount, respectively. The last digit represents the number of each profile in the plot.

layer.

4.2. Difference in preferential flow behavior for different forest restoration approaches

We carried out tracer experiments with different infiltration amounts for the two different forest restoration approaches to explore the characteristics of preferential flow behavior patterns. There was obvious preferential flow phenomena in each plot, where there was a distinct ‘finger’ flow in PF and ‘funnel’ flow in NSF. This can be explained by the mutability of texture and structure of shallow soil in the vertical direction (Wang and Zhang, 2017). In addition, due to the different forest restoration approaches, the rotten roots in the root layer were also different. After the succession of NSF, the rotten roots in the surface soil were higher than PF. Since root death could produce organic acids, amino acids, enzymes, sugars and other mixed organic matter, which can change soil structure, increase soil porosity, and thus enhance water infiltration capacity (Ghestem et al., 2011). This is one of the reasons that NSF produces wider paths ‘funnel’ flow. The root soil layer

characteristics of NSF were conducive to improving the horizontal and vertical connectivity of soil pores, thus increasing the width of water flow path. We further tested the spatial variability of each site preferential flow. A value results (Table 3) showed that NSF preferential flow variability was higher than for PF, especially in the horizontal direction. Spatial autocorrelation was high, which is mainly influenced by environmental structural factors (such as parent matter, vegetation, etc.) (Luo et al., 2019). Meanwhile, DC changed to a varying degree with the increase of soil depth, indicating that environmental structural factors have a strong interaction with the preferential flow patterns (Ding et al., 2023). In addition, increasing infiltration amount can not only increase the horizontal and vertical connectivity of the preferential flow channel. Fig. 5 (a)–(l) shows, that with the increase of infiltration amount, the soil depth of DC with sharp divergence also increased. When infiltration increases, the shallow soil layer preferential flow will gradually extend horizontally until it becomes matrix flow and preferential flow interact region, and more water will infiltrate into the deep soil in the form of matrix flow, thereby reducing the preferential flow development (Fig. 10). That is to say, the increase of infiltration plays an important

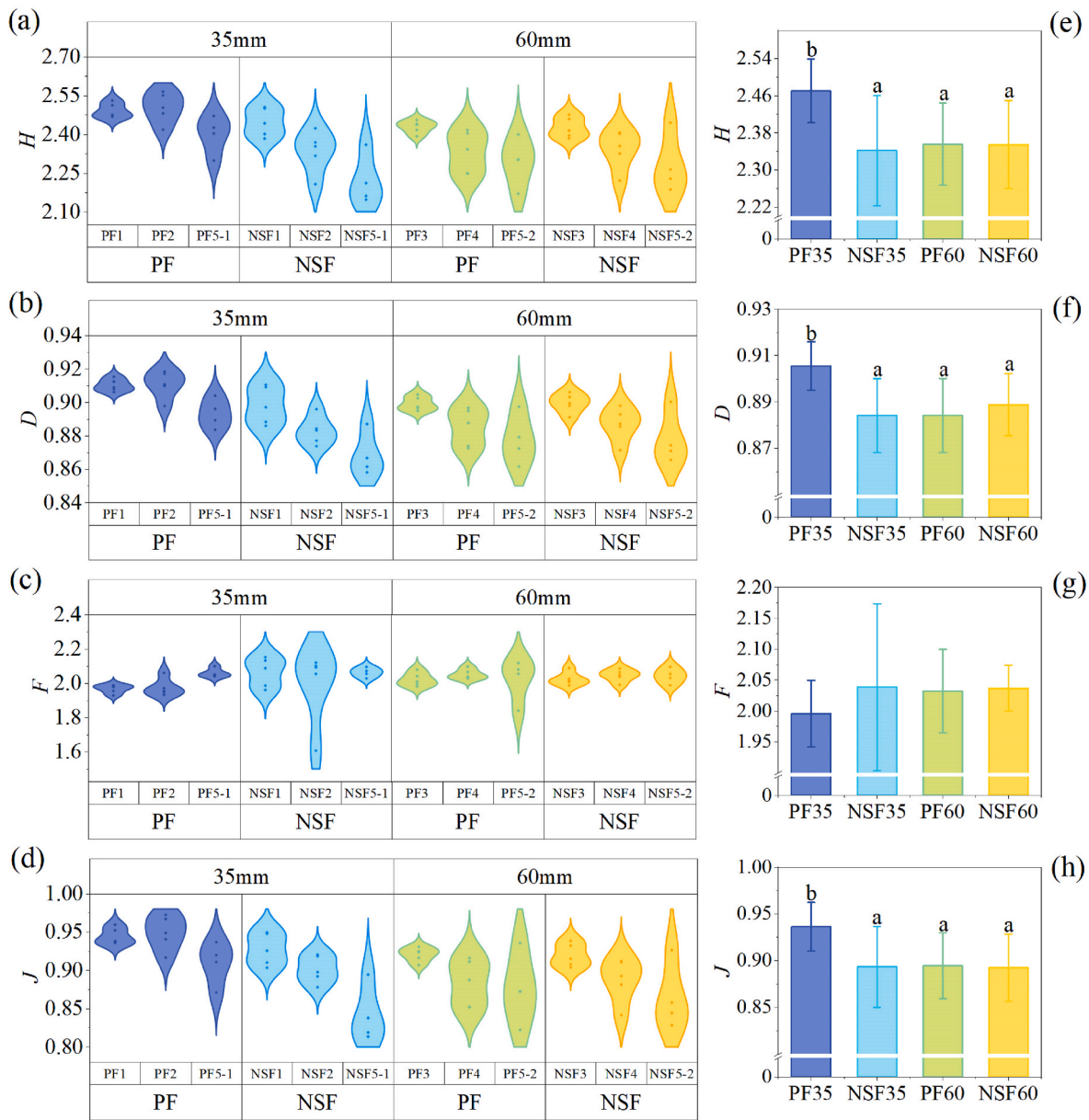


Fig. 9. Preferential flow pathway variation for two different forest restoration approaches. (a)–(d) violin plot of preferential flow pathways distribution (solute distribution) diversity characteristics in each plot, (e)–(h) comparison of preferential flow pathways distribution (solute distribution) diversity parameters (mean $\pm$ SE, $n = 5$ or 4) among different forest restoration approach under different infiltration amount. PF, plantation forest; NSF, natural secondary forest; under 35 mm and 60 mm infiltration amount, respectively. Bars with different lower-case letters (a and b) indicate significant difference among different sites at $p < 0.05$.

Table 5

Mean value, mean square deviation, and weight coefficient of each parameter in the design of *PFI*.

Preferential flow parameter	Mean value	Mean square deviation	Weight coefficient
<i>DC</i>	0.498	0.261	0.126
<i>CV</i>	0.613	0.224	0.109
<i>PF-fr</i>	0.757	0.223	0.108
<i>UniFr</i>	0.757	0.223	0.108
<i>Li</i>	0.514	0.230	0.111
<i>D</i>	0.557	0.257	0.124
<i>H</i>	0.558	0.252	0.122
<i>F</i>	0.770	0.151	0.073
<i>J</i>	0.570	0.247	0.120

role in the change of preferential flow development, which is consistent with the conclusion of Guan et al. (2023) on the dye tracer test in Karst forestland. Overall, the increase in infiltration significantly changed *DC* and *CV* ($p < 0.01$), *UniFr*, and *PF-fr* ($p < 0.05$) in each forest type. Under the same rainfall conditions, except for the *UniFr* and *PF-fr*, significant difference ($p < 0.05$) at 60 mm infiltration was observed, there was no significant difference in *DC*, *UniFr*, *PF-fr*, *CV* and *LI* between PF and NSF ($p > 0.05$). These results indicate that there were some differences in the preferential flow development between the two forest types. Combined with the results of each index, *UniFr* is an index to characterize the depth of matrix flow, and it is also a depth index which preferential flow alone begins to differentiate. The results showed that no matter what kind of infiltration conditions, the preferential flow began to diverge earlier in PF. *PF-fr* characterizes the proportion of preferential flow in the vertical profile, indicating that more infiltrated water was transported through preferential flow channels in PF. *LI* quantified the wetting front of

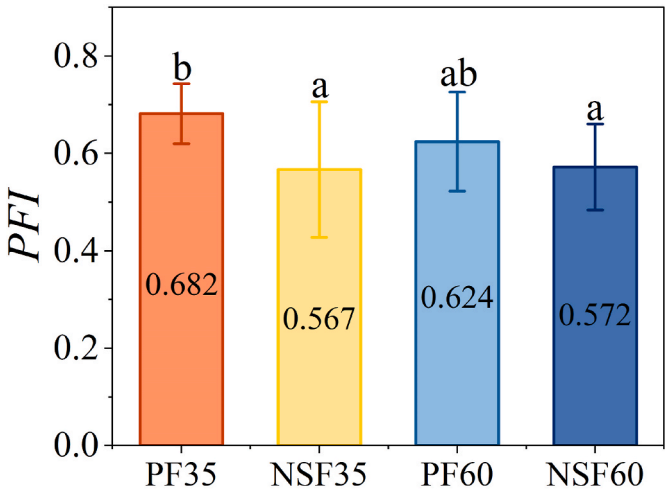


Fig. 10. Comparison of evaluation index (*PFI*) (mean $\pm$ SE, $n = 14$) among the two forest restoration approaches. PF, plantation forest; NSF, natural secondary forest; 35 and 60, under 35 mm and 60 mm infiltration amount, respectively. Bars with different lower-case letters (a and b) indicate significant difference among different sites at $p < 0.05$.

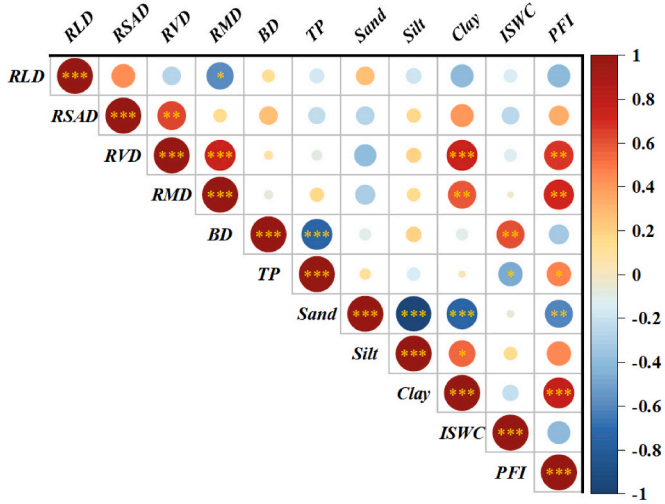


Fig. 11. Correlation analysis of preferential flow development (*PFI*) with soil physical properties and root traits factors. *PFI*, preferential flow index; *ISWC*, initial soil water content; *Clay*, clay content; *Silt*, silt content; *Sand*, sand content; *TP*, total porosity; *BD*, bulk density; *RMD*, root mean diameter; *RVD*, root volume density; *RSAD*, root surface area density; *RLD*, root length density. * indicate significant differences at $p < 0.05$, ** indicate significant differences at $p < 0.01$, *** indicate significant differences at $p < 0.001$.

preferential flow in soil profile, which is an index to characterize the non-equilibrium of preferential flow (Ding et al., 2023). There was no significant difference in *LI* between the two plots, but *PF* was higher than *NSF*, indicating that *PF* had greater non-equilibrium degree of

preferential flow and more preferential flow paths. That is, under different forest restoration types, the preferential flow of *PF* diverged relatively early and the degree of non-equilibrium is greater, with preferential flow dominating the infiltration process.

Preferential flow is a form of infiltration in which water carries solutes preferentially and rapidly downward in soil. Therefore, in addition to paying attention to the development characteristics of water flow morphology, we also studied the migration characteristics of solutes through dual tracers. Our previous studies have shown that preferential flow affects the redistribution of soil water content in soil profiles and leads to differences in the spatial distribution of soil water in the study area (Wang et al., 2024). In this study, the migration of solute also showed the same pattern consistent with the distribution of water flow (*DC*). That is, both the soil water content and the solute concentration in the soil profile increased to different degrees after infiltration, and the increase rate decreased along the soil profile. These results indicate that gravity is the driving factor for the downward migration of water and solute (Ding et al., 2023). Interestingly, we compared the solute Br^- relative concentration and *DC* (Figs. 7 and 8) in the same profile and found that the solute Br^- diffusion range in the soil was larger than the observed staining path range. This is mainly since the solute tracer NaBr we selected is colorless, and its path cannot be directly observed in the profile flow patterns. In addition, as the essence of brilliant blue is organic matter, it easily adsorbs to the soil surface and organic matter-rich areas (Ketelsen and Meyer-Windel, 1999). Thus, NaBr has stronger migration ability than brilliant blue, which is consistent with findings by Bogner et al. (2008). Therefore, preferential flow parameters based solely on profile staining do not fully characterize the path characteristics of preferential flow. In this study, we evaluated the diversity characteristics of preferential flow paths through the solute migration diversity index on the profile. Combined with the morphological characteristics of preferential flow, we found that the flow path has the characteristics of connectivity and isolation. Connectivity is reflected in the interconnected macropore grid of soil structures, while the presence of isolated paths increases the heterogeneity of macropore distribution (Zhang et al., 2018), and increases the diversity of paths. In this study, we introduced α -biodiversity theory to evaluate the distribution characteristics of water flow pathways in different sites. In terms of diversity indices (*D*, *H*), *PF* under 35 mm infiltration condition had highest preferential flow pathway diversity, with both the abundance and complexity of preferential flow pathways being higher than those in other sites. However, there was no significant difference in the richness index (*F*) of preferential flow pathways among the sites, indicating a tendency towards uniformity in the types of preferential flow pathways in different sites. Therefore, the diversity difference of preferential flow pathways among the sites was mainly due to differences in pathway abundance and uniformity. From the perspective of uniformity (*J*), the *J* value in *PF* was higher, primarily because the sand content in the *PF* soil was lower compared to that in *NSF* (Table 4), leading to a more stable spatial distribution pattern of solute concentrations (Jacobs et al., 2004), i.e., higher uniformity of preferential flow pathways. Some studies have also pointed out that *BD* (Cosh et al., 2008) and *ISWC* (Williams et al., 2020) are important factors leading to differences in the spatial distribution of solute concentration. In addition, the spatial heterogeneity of gravel distribution at each site affects the water infiltration path (Wang et al., 2024) and increases the number of macropores

Table 6
Path analysis results of factors affecting preferential flow development (*PFI*).

Parameter	Direct path coefficient	Indirect path coefficient				Sum of coefficient
		Clay	TP	RVD	Total	
Clay	0.463		0.018	0.343	0.361	0.824
TP	0.487	0.019		−0.044	−0.025	0.462
RVD	0.378	0.280	−0.034		0.246	0.624

Note: *Clay*, clay content; *TP*, total porosity; *RVD*, root volume density.

for effective water and solute transport (Baetens et al., 2009).

4.3. Impact factors on preferential flow development degree

We estimated the preferential flow development through the preferential flow paths and diversity index. We explored the factors influencing the preferential flow development, including soil physical properties and root traits. Of these, the effects of clay content, sand content, *TP*, *RMD*, and *RVD* for *PFI* were statistically significant. Further analysis showed that clay content, *TP* and *RVD* were the main factors affecting the preferential flow development. The results of Schaik (2010) indicated that soil texture was an important spatial parameter for predicting preferential flow. Wang et al. (2018) also noted that soil with high clay content reduce macropore connectivity in the horizontal direction and enhance preferential flow compared to soil with high sand content, which is consistent with our findings. Water transport between soil macropores and soil matrix is a critical process during water infiltration (Jiang et al., 2015). The water exchange between the macropores of the surface soil and the soil matrix is the largest (Zhang et al., 2018), so the infiltration of the surface soil exhibits matrix flow and preferential flow interact strongly dominance. With soil depth, soil *TP* decreases, water exchange between the macropores and the soil matrix weakens (Wu et al., 2022), and water infiltration pattern appears to be dominated by preferential flow. As the *TP* of NSF was higher than that of PF, NSF had a wider range of matrix flow and preferential flow interact strongly regions. In addition, vegetation species within different forest restoration stands have differences in root morphology characteristics that induce differences in water infiltration patterns (Yu et al., 2016), and macropores generated by roots can serve as preferential flow paths to enhance soil pore structure connectivity and permeability (Jiang et al., 2018; Guo et al., 2019).

It was found that the preferential flow development for PF was higher than that of NSF under the conditions of heavy rainfall (35 mm) and extreme rainfall (60 mm). In addition, increasing rainfall can reduce the preferential flow development, which is similar to previous findings (Wu et al., 2022). Notably, although preferential flow for PF was highly developed, it did not mean that it had the greatest intensity resulting in high infiltration. As can be seen in Fig. 4, the flow was stronger in all directions in the NSF and thus the infiltration profile was more connected, which reduced the preferential flow. Our previous studies found that NSF had stronger infiltration capacity (Wang et al., 2024). Compared to PF, NSF has deeper regions where matrix flow and preferential flow interact more strongly. In other words, PF had a smaller range of channels for transporting water and solutes, and more water carrying solutes moved down through these channels, increasing the potential for subsurface erosion to occur. Therefore, soil erosion management in the study area should be considered in the integrated surface and subsurface hydrological processes. As mentioned above (section 4.1), soil erosion was more prone to occur on the NSF surface, and that PF was more at risk of soil erosion in the subsurface when combined with the water infiltration process. Therefore, the forestland management strategy in the study area should be considered to increase the ecological sustainable development potential of the forest by appropriately increasing artificial establishment measures to natural secondary forests and managing extensive in plantation forests.

5. Conclusions

This study investigated the effects of different forest restoration approaches on soil physical properties, root traits, and preferential flow patterns and development in a paleo-periglacial landform. The main conclusions are as follows.

- (1) The dye experiments showed obvious preferential flow for PF and NSF in the study area, where the flow in PF was of 'finger' shape and NSF of 'funnel' shape. In addition, the horizontal preferential

flow path spatial variability was strongest for NSF. Preferential flow diverged earlier, was more non-equilibrium, and developed to a higher degree for PF than for NSF. Besides, preferential flow was the dominant form of infiltration for water transport into deeper soil in the study area.

- (2) Increasing the infiltration amount enhanced the connectivity of the water flow channels and reduced the preferential flow development.
- (3) The dual-tracer tests showed that solute Br^- diffusion was more extensive than Brilliant Blue stained water flow pathways. We quantified the preferential flow development (*PFI*) through a combination of preferential flow morphology indices and pathway diversity (solute distribution) indices.
- (4) Differences in the *PFI* can be explained by differences in clay content, *TP*, and *RVD*, with total path coefficients of 0.824, 0.462, and 0.624, respectively.

In addition, the results of this study showed that the soil surface of the NSF was more susceptible to soil erosion, and although PF establishment enhances surface soil stability, PF was at a higher potential risk of erosion in the subsurface. Therefore, the management strategy of forestland should be considered to increase soil erosion prevention and control measures in NSF and enhance the management of near-natural patterns in PF, to enhance the ecological sustainability of forestland in the study area. Furthermore, further research needs to be carried out for in-depth revelation of how vegetation in different forest restoration stands interacts with soil and influences the preferential flow patterns. This will also facilitate the hydrological management of different water conservation forest types to cope with water supply and demand stress in the region.

CRediT authorship contribution statement

Di Wang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jianzhi Niu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Zhengyu Dai:** Supervision, Software, Methodology, Investigation. **Tao Yang:** Methodology, Investigation, Formal analysis. **Yubo Miao:** Visualization, Validation, Data curation. **Linus Zhang:** Writing – review & editing, Supervision, Software. **Xiongwen Chen:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization. **Ronny Berndtsson:** Writing – review & editing, Validation, Supervision, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was funded by the National Key Research and Development Program of China (2022YFF1300804), the National Natural Science Foundation of China (41877154).

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